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October 31, 2026

Editor-in-Chief
IEEE Access
IEEE Publishing Operations

Dear Editor,

We submit the manuscript "Less a Weak Source than a Different One: Vulnerability, Similarity and Direction in CNN-ViT Adversarial Transfer under an Architecture-by-Initialization Design" for consideration as a Regular Article in *IEEE Access*.

Adversarially trained Vision Transformers often look like weak transfer sources against CNNs: examples crafted on the ViT fool a CNN less often than examples crafted on the CNN fool the ViT. With one CNN and one ViT this asymmetry cannot be divided into a source part and a target part, so the common reading that the ViT crafts weaker examples has so far rested on a quantity that cannot decide it. The manuscript separates the two by factoring each log target-correct transfer rate into the target's white-box vulnerability, a symmetric pairwise similarity and a directional residual, and by averaging the last two over a fixed panel of seven robust CNNs from RobustBench. The title states the answer in quantitative form. On CIFAR-10, a share of 0.936 (90% interval $[0.925, 0.948]$) of ViT-Tiny's panel source gap relative to ResNet-18 lies in the symmetric term, where a pure source effect would give one half, and the ViT's directional term ($+0.012$) is equivalent to zero within the registered bound of $\pm 0.10$. Most of what looks like a weak source is dissimilarity that runs in both directions.

The study was designed so that this reading could fail. The original pair (three training runs per architecture, a validation split fixed before training, per-sample records on the full test set) is extended by six new runs that complete a two-by-two design crossing architecture with initialization, random or ImageNet, on a matched input pipeline and training recipe. Predictions, tests and decision rules were registered in advance: the main registration was frozen on 29 October 2026 before any of these runs, and an addendum covering CIFAR-100, ImageNetV2 and gradient alignment on 30 October 2026 before any of its matrices was computed. Every registered test ran through one frozen analysis script whose SHA-256 digest is recorded in each output. The manuscript also states what the registration does not cover. The CIFAR-10 panel tests extend a first seed that had already been seen, the architecture prediction was written with the pair's similarity gap known, the refutation rule for the ViT's directional term was widened to both signs after data had been seen, and one registered sensitivity analysis, a hierarchical bootstrap that the frozen script lacked, was computed by a separate script after the results and an internal review had been seen. Analyses outside the registration are labelled exploratory.

The first contribution is an identifiability result. In a two-model design the asymmetry identifies only a difference of potentials, and an additive source-plus-target fit in a family with a single ViT can load the similarity among the CNNs onto the ViT's source term. A fixed external panel combined

with a multiplicative decomposition makes vulnerability, similarity and direction identifiable for each model. The decomposition is an accounting identity, related to the quasi-symmetry model of Caussinus and to the transferability analysis of Demontis et al.; we do not present it as a finding, and the paper's content lies in the values its terms take and in how they respond to an intervention.

The second contribution is the measured size of these terms on two datasets. Besides the symmetric share and the directional term above, ViT seeds resemble each other more than they resemble the panel (+0.566 on the log similarity scale). On CIFAR-100 the symmetric share (0.840) and the seed similarity replicate, but the registered equivalence of the ViT's directional term is refuted (+0.143), and it is refuted under the original one-sided rule as well; on that dataset the ViT is also a directionally weaker source. The manuscript reports this refutation in the abstract and does not soften it.

The third contribution is an intervention on the pair itself. Trained from scratch on the same pipeline, the ViT is less similar to the panel than a ResNet-18 ($-0.364$, with a 90% interval of $[-0.415, -0.314]$); this contrast moves together with the ViT's lower clean accuracy (73.53% against 85.73%), which the design cannot separate from architecture, and the paper says so in the abstract. ImageNet initialization raises the ViT's panel similarity more than the ResNet's (+0.260), and, averaged over the two initializations, architecture leaves the log-scale directional term within the margin (+0.003); all three registered tests are supported after Holm correction. The gap is retained, at a fraction of 0.960, when each source uses the best of PGD-10, momentum PGD and our adaptation of token gradient regularization, which transferred worse than PGD-10 from every ViT source, and the registered AutoAttack rule flags none of the six new models for gradient masking.

The fourth contribution concerns the measurement layer underneath. With models, data and attack held fixed, the conditioning protocol alone moves the asymmetry of the pair from +4.36 to +19.37 points (mean per-pair spread $15.01 \pm 0.84$ points), whereas independent training runs move it with standard deviations of 0.23 to 1.27 points within a protocol. Unconditioned and target-correct rates are linked by the law of total probability through a premise that measures at least 0.989 on our adversarially trained targets, and on the scale $h = -\log(1 - T)$ the two protocols differ only in the target potentials, up to a small adjustment term. This is why the decomposition is computed on one stated protocol.

The claim has a definite boundary, and the abstract, introduction and conclusion state it in the same terms. A small directional term holds on the log scale, against a CNN panel and on CIFAR-10 only: on the logit scale the ViT's directional term is strongly negative, against a single transformer reference the scratch-trained ViT is a directionally weaker source, and on CIFAR-100 the equivalence is refuted. The architecture contrast compares whole packages, patch stem, attention and parameter count together. The ImageNetV2 evidence is descriptive and not a test of the two-by-two design: in two same-recipe CNN-transformer pairs the transformers are less similar to the panel ($-0.142$, 90% image-bootstrap interval $[-0.197, -0.090]$), and the only scratch-trained ImageNet pair gives an uncertain estimate. Input-gradient cosine tracks pairwise similarity on three datasets (Spearman correlation 0.893 to 0.944 after additive model effects are removed); we read this as an association that a gradient-alignment account requires, not as evidence of a mechanism, and one ImageNet pair combines a near-zero cosine with above-median similarity. None of this concerns the absolute ordering, which favors the CNN under every attack (AutoAttack accuracy $37.93 \pm 0.14$% against $29.14 \pm 0.40$%).

We believe the manuscript suits *IEEE Access* because its subject is how a widely reported quantity should be read, and the reporting requirements it closes with (state the conditioning set, measure

target vulnerability separately from similarity, use models outside the compared pair, name the scale on which direction is read) apply to any study that compares adversarial transfer across model families. Every number in the text is recomputed from per-sample records and checked against the released artifacts by scripts that are part of the release.

A preliminary version of this study, limited to CIFAR-10 and to the ResNet-18 and ViT-Tiny pair, has been submitted to ATEEC 2026 and is under review; the manuscript cites it and describes the overlap in its introduction. That version reports the conditional decomposition, the four-protocol comparison and the gradient and drift measurements. The journal version adds the vulnerability-similarity-direction decomposition with its fixed panel, the architecture-by-initialization design with six new training runs, the attack-side and masking checks, the CIFAR-100 and ImageNetV2 panels, the gradient-alignment test, the protocol identity with its measured premise and the registration of predictions before measurement. It also corrects one claim of the conference version: the direction of the asymmetry is not stable once the two architectures are matched in clean accuracy. We will declare the conference paper in the portal's prior-publication field and update its citation if the conference decision arrives during review.

As stated in the Acknowledgment of the manuscript, the authors used Claude (Anthropic) to assist with drafting and revising the text and with implementing the analysis and verification scripts. The study design, the experimental protocol and all conclusions are the authors', who verified every reported number against the released artifacts. All figures are produced by plotting scripts or, for the schematics, drawn as vector graphics; none is produced by an image-generation model.

The manuscript is original, has not been published elsewhere and is not under consideration by any other journal. Both authors have read and approved the submitted version and declare no conflict of interest. All datasets and RobustBench checkpoints used are public, and the analysis pipeline, per-sample records, registration documents, frozen analysis script and cell tables are available at `https://github.com/cagrisahin58/cnn-vit-robustness-protocols`; trained checkpoints are available from the authors on request.

Thank you for considering our manuscript.

Sincerely,

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